

NON-LETHAL SAMPLING OF FISH TISSUES FOR GENETIC ANALYSIS
STANDARD OPERATING PROCEDURE (IACUC Assigned SOP #4)

March 27, 2008

By

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A. SCOPE AND APPLICABILITY

This method is applicable to non-lethal field sampling fish for subsequent genetic analysis. This protocol is non-taxon specific and, as such, is applicable to nearly all species of fish. The methods and approaches used in this SOP have been evaluated and employed by numerous laboratories across the world and are consistent with the AFS/AIFRB/ASIH Guidelines for the Use of Fish in Research (2004).

B. SUMMARY OF METHOD

Fish are sampled live from their natural environment using a variety of established methods including but not limited to fyke-nets, trap nets, seines, and electrofishing (Hayes et al. 1996; Hubert 1996; Reynolds 1996). When appropriate, fish are measured for length and weight (Anderson and Neumann 1996) followed by the sampling of tissue (Kelsch and Shields 1996; Jennings et al. In Press). Tissue choices include the removal of a single, small (<10 mm) clip of the distal end of a fin (Note: choice of fin will take into account the biological impact of sampling certain fins in different taxa; this is generally the pelvic, anal, or caudal fin) and/or the removal of 5-10 scales from the dorsal side of the lateral line approximately half-way down the fish (DeVries and Frie 1996). Fin clips are placed in a pre-labeled screw-top 2.0 mL tube and preserved in 95% non-denatured ethanol. Scale samples can be placed in pre-labeled coin envelopes between the pieces of a piece of filter paper. Samples are then taken to the genetics lab for analysis.

C. HEALTH AND SAFETY WARNINGS

This SOP is designed to be performed in a field environment. As such, all appropriate precautions should be taken. These include but are not limited to the use of appropriate clothing including gloves for safe handling of fish, proper training (all individuals using and operating a boat through the Cooperative Fishery Research Unit are required to undergo USGS boat safety training and have up-to-date Red Cross First Aid and CPR training), and sound judgment (e.g., if weather is too rough to safely operate and sample specimens, work should be delayed).

In addition, specific health and safety precautions include being careful with the sharp scissors/shears used to clip fins, the knife/scalpel employed in scraping scales, and any knives used for euthanasia. Standard equipment for all sampling excursions should include a first aid kit and a means of communication (i.e., cell-phone, walkie-talkie, etc.).

Chemicals used in this SOP are 95-100% non-denatured ethanol, and a Tris-HCl buffered tricaine methane sulfonate (MS-222®) solution. The following bullets outline key safety issues to apply when working with these chemicals. Please note an MSDS for all chemicals are found in the Biology Stockroom (TNR Room 193) and in the Molecular Conservation Genetics Laboratory (TNR Room 340). Individuals should familiarize themselves with all risks and procedures associated with any hazardous material before handling and using.

◆ Ethanol (non-denatured, 95-100%)

CAS#: 64-17-5

Flammable liquid and vapor. Keep away from open flames and sparks.
Wear appropriate gloves and goggles
When possible have the ethanol pre-aliquoted into prelabeled tubes or use a squirt bottle when you return to shore to add ethanol to samples.

◆ **Tricaine methane sulfonate**

Only take premixed stock solutions into the field (see recipes later in SOP).
Wear gloves and use a utensil to dilute stock solution in tank.
Wear gloves to handle animals euthanized in tricaine methane sulfonate.
Dispose of excess solution by flushing down the drain to a sanitary sewer with an excess of water. If in a remote location where a sewer is unavailable, further dilute solution with water and dump waste on land away from water.
Do not discard directly into surface water, storm water conveyances or catch basins

D. EQUIPMENT AND SUPPLIES

-All supplies available upon request from the Molecular Conservation Genetics Lab in TNR 340.

Fin Clips:

- 1) Pre-labeled screw-cap tubes (must include a rubber insert to avoid evaporation of ethanol)
- 2) Tube rack holding tubes in a prearranged order
- 3) Tissue Scissors or snips (must be sharp to allow a single cut to sever the sample)
-have at least two pairs
- 4) Forceps (at least two pairs)
- 5) Clipboard
- 6) Data Entry Sheet
- 7) Pencil/Write-In Rain Pen
- 8) Squeeze Bottle (250-500 mL) filled with 95% Ethanol
- 9) Fish Measuring Board
- 10) Electronic balance and tub
- 11) Sharp 'boning' knife
- 12) Blunt instrument for pithing (dissecting probe, screwdriver, etc.)
- 13) Euthanasia/anesthesia 'tub' filled with final working solution of at least 500 mg/L MS-222® (1:4 dilution of stock solution with dH₂O; modified from Cornell University Fish and Amphibian Euthanasia SOP at:
www.research.cornell.edu/care/documents/SOPs/CARE306.pdf)

MS-222® Stock Solution (1 L)

2g tricaine methane sulfonate
21 mL 1M Tris-Cl (pH 9)
960 mL ddH₂O

-Adjust pH to 7.0 and bring to 1 L final volume with dH₂O
-Store in a dark bottle for no longer than 15 days

Scale Sampling:

- 1) Pre-labeled coin envelopes with filter paper insert arranged sequentially in a box
- 2) Pocket knife and/or scalpel
- 3) Forceps (at least two pairs)
- 4) Clipboard
- 5) Data Entry Sheet
- 6) Pencil/Write-In Rain Pen
- 7) Fish Measuring Board
- 8) Electronic balance and tub
- 9) Euthanasia 'tub' filled with working MS-222® solution (~500 mg/L; see recipe above)

E. SAMPLE COLLECTION, PRESERVATION, HANDLING, AND SHIPPING

Fin Clips:

- 1) Organize work space to maximize fish handling efficiency. It is imperative the fish is handled in a short period of time. Total procedure should not take more than 1-2 minutes.
- 2) Collect morphological data (i.e., length, weight, etc...).
- 3) Cut a "nickel" size piece of fin tissue usually from the target fin using scissors.
- 4) Using forceps, place tissue in the labeled screw cap tube (Note: place tissue in tubes in consecutive order beginning with the smallest number to minimize confusion).
- 5) Fill tubes with ethanol and screw on cap securely to prevent the ethanol from evaporating (Note: if handling a large number of fish, you can wait for a pause in sampling to add ethanol; just make sure the lids are put back on the tubes to prevent mixing tubes and lids).
- 6) Place tubes in tube rack in sequential order.
- 7) Record tube number on the data sheet so tissue samples can be matched up with morphological data collected for each individual.
- 8) Rinse scissors and forceps in water between samples to minimize contamination risk (Note: lake/river water is sufficient; no visible blood, 'slime', or tissue should be present between samples).
- 9) To prevent the accidental spread of disease and/or pathogens, use new snips/scissors and forceps for each new population. The used snips/scissors and forceps can be autoclaved to decontaminate.
- 10) Label tube box with site specific information (location, date, range of sample numbers, and name of individuals collecting sample).

Scale Sampling:

- 1) Organize work space to maximize fish handling efficiency. It is imperative the fish is handled in a short period of time. Total procedure should not take more than 1-2 minutes.
- 2) Collect morphological data (i.e., length, weight, etc...).
- 3) Scrape approximately 5-10 scales from the side of the fish with a clean pocketknife or scalpel. .
- 4) Put scales (on knife blade) into a prelabeled scale envelope between the folds of a piece of filter paper. Scrape off scales and close envelope.
- 5) Place envelope in back of box (keep order consistent).
- 6) Record envelope number on the data sheet so scale samples can be matched up with morphological data collected for each individual.
- 7) Rinse knife in water between samples to minimize contamination risk (Note: lake/river water is sufficient; no visible blood, 'slime', or tissue should be present between samples).
- 8) To prevent the accidental spread of disease and/or pathogens, use new knives/scalpels for each new population.

Injured fish/Euthanasia

(NOTE: partially adopted from University of Texas at Austin, Appendix E: IACUC Guideline: Humane Euthanasia of Laboratory Animals, available at: <http://www.utexas.edu/research/rsc/animalresearch/policies/appendix/e.php>).

-All reasonable efforts shall be made to consistently apply the principles and practices of euthanizing animals including the 10 Principles for Animal Euthanasia established by Demers et al. 2006. The following methodology is consistent with the recommendations of AFS/AIFRB/ASIH (2004) and CCAC (2005).

- 1) If a fish becomes injured severely enough to question the survival of the individual, this specimen should be euthanized.
- 2) All effort should be made to minimize the suffering of the organism.
- 3) Place the individual in the euthanizing/anesthetizing solution (see ingredients above).

- 4) For some fish, death is rapid (<15 minutes) in MS-222® and is determined by the lack of opercular movement for at least 5 minutes. For these fish, proceed to step 7.
- 5) For all fish not anesthetized rapidly in concentrated MS-222®, when fish is anesthetized, remove from solution and decapitate using a sharp knife.
- 6) Following decapitation, cranially pierce the organism to disrupt consciousness and cause death.
- 7) Fish is euthanized and should be discarded in a landfill.

F. JUSTIFICATION

“It is becoming increasingly common to remove small pieces of fin tissue to obtain DNA for sequencing and other types of molecular studies. Pieces of fin preserved in 70% or higher concentrations of ethanol can yield adequate amounts of DNA, and the fin clips are not harmful to the fish. This technique is especially important when working with imperiled species and small populations,” AFS/AIFRB/ASIH (2004).

The use of non-lethal sampling has resulted in remarkable benefits to the fisheries and conservation biology field in terms of genetic and contaminant analyses. This approach minimizes the impact on the fish population and aquatic community as a whole in terms of not significantly impacting annual mortality and/or community structure and function. The occurrence of literally billions of copies of DNA in a fin clip no larger than 10 mm² or in 3-10 scales represents a vast improvement in terms of animal care and use compared to the whole body analyses necessary just 15-20 years ago in studies reliant on protein polymorphisms.

Minimizing stress on the organism during this procedure focuses on the question of whether or not to use an anesthetic and the time spent handling the fish. The use of anesthetics such as tricaine methane sulfonate (hereafter referred to as MS-222® but also known as Finquel®, TRICAINES, and TMS) in this procedure is problematic. MS-222® is a suspected carcinogen. This exposes the researcher(s) and field personnel to a hazardous material while often being in the bow of a boat sampling fish populations in inclement weather. Even in the best of weather, the water is often choppy and the risk to the field personnel is great that some exposure to the chemical will occur. Secondly, the fish most commonly sampled by the researcher(s) are gamefish that are targeted for consumption by anglers. There is an FDA mandated 21-day withdrawal time necessary for all fish anesthetized by MS-222® that would be consumed (AVMA 2007). In all systems that the researchers work, the assumption these fish *could* be consumed within this period is a realistic possibility. This would mandate a quarantine or holding-period of 21 days for all sampled fish. This would undoubtedly introduce more stress to the organism than rapid handling time alone with no anesthetic. Since the partial clipping of a fin and/or removal of scales is categorized as a temporary pain with no lingering effect (Guy et al. 1996), we believe the use of MS-222® and related drugs is inappropriate for non-lethal field sampling of fish tissues. Appropriate preparation and organization on the researcher's part should minimize the handling time from the sampling gear to the return of the fish to the water to

no more than 1-2 minutes consistent with AFS (2004) thus resulting in sufficient reduction in stress.

MS-222® can and should be used as either a sole euthanizing agent (when appropriate) or as part of a 3-step euthanasia approach if a sampled fish has experienced significant injury and the field personnel determine the probability of its survival is low to zero. In the sampling gear commonly employed for fisheries research in the Midwest, this is a relatively rare occurrence based on personal observation. However, in cases where injury does occur, the specimen can be submerged in a concentrated solution of MS-222® that is 500 mg/L and in a sealed container (AFS/AIFRB/ASIH 2004; CCAC 2005; AVMA 2007).

G. REFERENCES

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